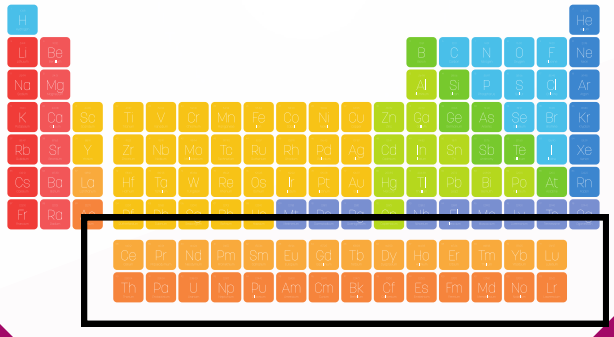
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Inorganic Chemistry

ENTHUSIAST COURSE



STUDY MATERIAL

f - Block Element

English Medium

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f-BLOCK COMPOUNDS

Inner Transition Elements :

The elements in which the additional electron enters in (n – 2)f orbitals are called inner **transition elements or f-block elements.**

Position in the periodic table :

The lanthanides resemble with Yttrium in most of their properties. So it became necessary to accomodate all the fifteen elements together at one place. This has been done by placing the first element, lanthanum below yttrium and placing the remaining fourteen elements separately in the lower part of the periodic table.

Lanthanide series (Z = 58 – 71) (Ce – Lu)

Actinide series (Z = 90 – 103) (Th – Lr)

LANTHANIDES (RARE EARTHS OR LANTHANONES)

(i) Lanthanides are reactive elements so do not found in free state in nature.

(ii) Most important minerals for lighter Lanthanides are – Monazite, cerites and for heavier lanthanides – Gadolinite and Xenotime

Electronic configuration :

(i) The general configuration of lanthanides may be given as $4f^{1-14}5s^25p^65d^{0-1}6s^2$.

Atomic Number	Element	Symbol	Outer electronic configuration	
			Atomic	+3 ion
58	**Cerium**	**Ce**	**$4f^1\ 5d^16s^2$**	**$4f^1$**
59	Praseodymium	Pr	$4f^3\ 6s^2$	$4f^2$
60	Neodymium	Nd	$4f^4\ 6s^2$	$4f^3$
61	Promethium	Pm	$4f^5\ 6s^2$	$4f^4$
62	Samarium	Sm	$4f^6\ 6s^2$	$4f^5$
63	Europium	Eu	$4f^7\ 6s^2$	$4f^6$
64	**Gadolinium**	**Gd**	**$4f^7\ 5d^1\ 6s^2$**	**$4f^7$**
65	**Terbium**	**Tb**	**$4f^9\ 6s^2$**	**$4f^8$**
66	Dysprosium	Dy	$4f^{10}\ 6s^2$	$4f^9$
67	Holmium	Ho	$4f^{11}\ 6s^2$	$4f^{10}$
68	Erbium	Er	$4f^{12}\ 6s^2$	$4f^{11}$
69	Thulium	Tm	$4f^{13}\ 6s^2$	$4f^{12}$
70	**Ytterbium**	**Yb**	**$4f^{14}\ 6s^2$**	**$4f^{13}$**
71	**Lutetium**	**Lu**	**$4f^{14}\ 5d^1\ 6s^2$**	**$4f^{14}$**

(ii) It is to be noted that filling of 4f orbitals in the atoms is not regular. A 5d electron appears in gadolinium (Z = 64) with an outer electronic configuration of $4f^75d^16s^2$ (and not $4f^86s^2$). This is because the 4f and 5d electrons are at about the same potential energy and that the atoms have a tendency to retain stable half filled configuration.

(iii) On the other hand, the filling of f-orbitals is regular in tripositive ions.

(iv) After losing outer electrons, the f-orbitals shrink in size and became more stable.

(v) **Pm** is the only synthetic radioactive lanthanide.

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Oxidation states :

Lanthanides	Oxidation	Actinides	Oxidation state
Ce_{58}	+3, +4	Th_{90}	+4
Pr_{59}	+3, (+4)	Pa_{91}	(+4), +5
Nd_{60}	+3	U_{92}	(+3), (+4), (+5), +6
Pm_{61}	+3	Np_{93}	(+3), (+4), +5, (+6), (+7)
Sm_{62}	(+2), +3	Pu_{94}	(+3), +4, (+5), (+6), (+7)
Eu_{63}	+2, +3	Am_{95}	+2,(+3),(+4),(+5),(+6)
Gd_{64}	+3	Cm_{96}	+3, (+4)
Tb_{65}	+3, +4	Bk_{97}	+3, (+4)
Dy_{66}	+3, (+4)	Cf_{98}	+3
Ho_{67}	+3	Es_{99}	+3
Er_{68}	(+2), +3	Fm_{100}	+3
Tm_{69}	(+2), +3	Md_{101}	+3
Yb_{70}	+2, +3	No_{102}	+3
Lu_{71}	+3	Lr_{103}	+3

(Oxidation states in brackets are unstable states)

(i) The lanthanides contains two **s** electrons in the outermost shell, they are therefore expected to exhibit a characteristic oxidation state of +2. But for the lanthanides, the +3 oxidation is common.

(ii) This corresponds to the use of two outermost electrons ($6s^2$) alongwith one inner electron. The inner electron used is a 5d electron (in La, Gd and Lu), or one of the 4f electron if no 5d electrons present.

(iii) All the lanthanides attains +3 oxidation state and only **Cerium, Praseodymium,** and **Terbium** exhibit **higher oxidation state (+4). Eu and Yb** exhibit **+2** oxidation state.

(iv) Oxidation states +2 and +4 occur particularly when they lead to -

(a) A noble gas configuration **Ex.** Ce^{4+} (f^0). The formation of Ce^{IV} is favoured by its noble gas configuration, but it is a strong oxidant reverting to the common +3 state. The E° value for Ce^{4+}/Ce^{3+} is + 1.74 V which suggests that it can oxidise water. However, the reaction rate is very slow and hence Ce(IV) is a good analytical reagent.

(b) A half filled 'f' orbital **Ex.** Eu^{2+}, (f^7), Pr, Nd, Tb and Dy also exhibit +4 state but only in oxides, MO_2. Eu^{2+} is formed by losing the two s electrons and its f^7 configuration accounts for the formation of this ion. However, Eu^{2+} is a strong reducing agent changing to the common +3 state. Similarly Yb^{2+} which has f^{14} configuration is a reductant.

(c) A completely filled 'f' orbital **Ex.** Yb^{2+} (f^{14})

(v) Therefore, in higher oxidation state, they act as oxidising while in lower state as reducing agents.

Magnetic properties :

(i) In tripositive lanthanide ions the number of unpaired electrons regularly increases from lanthanum to Gadolinium (0 to 7) and then continuously decreases upto lutecium (7 to 0).

(ii) lanthanum and lutecium ions are diamagnetic, while all other tripositive lanthanide ions are paramagnetic. (Exception – Neodyomium is the most paramagnetic lanthanide).

(iii) Ce^{+4} and Yb^{+2} are also diamagnetic ions.

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Colour :

(i) The lanthanide ions have unpaired electrons in their 4f orbitals. Thus these ions absorbs visible region of light and undergo f–f transition and hence exhibit colour.

(ii) The colour exhibited depends on the number of unpaired electrons in the 4f orbitals.

(iii) The ions often with $4f^n$ configuration have similar colour to those ions having $4f^{14-n}$ configuration.

(iv) Lanthanide ions having $4f^0$, $4f^{14}$ are colourless.

Other Properties :

- All the lanthanoids are silvery white soft metals and tarnish rapidly in air.
- The hardness increases with increasing atomic number, samarium being steel hard. Their melting points range between 1000 to 1200 K but samarium melts at 1623 K.
- They have typical metallic structure and are good conductors of heat and electricity.
- Highly dense metals with high m.pts. do not show any regular trend.
- **Ionisation Energies :** Lanthanides have fairly low ionisation energies comparable to alkaline earth metals.
- **Electro positive character :** High due to low I.P.
- **Complex formation :** Do not have much tendency to form complexes due to low charge density because of their large size. Lu^{+3} is smallest in size can only form complex.
- **Reducing Agent :** They readily lose electrons so are good reducing agent.
- **Alloy :** Alloys of lanthanides with Fe are called Misch **metals**, which consists of a lanthanoid metal (~ 95%) and iron (~ 5%) and traces of S, C, Ca and Al.
- **Basic Nature :** $La(OH)_3$ is most basic in nature while $Lu(OH)_3$ least basic.
- **Carbide :** Lanthanides form MC_2 type carbide with carbon, which on hydrolysis gives C_2H_2.
- The lanthanide elements Eu and Yb dissolve directly in very high concentration in liquid ammonia.

Lanthanide contraction :

(i) In the lanthanide series with increasing atomic number, there is a progressive decrease in the size from lanthanum to lutetium or from La^{+3} to Lu^{+3}. This contraction in size is known as lanthanide contraction.

(ii) The general electronic configuration of these elements is $4f^{1-14}5s^2p^6d^{0-1}6s^2$. In these elements the added electron enters the deep seated f-orbitals and therefore experiences considerable pull by the nucleus.

(iii) Due to very poor shielding effect of (n-2)f electrons, they exert very little screening effect on the outermost $6s^2$ electrons.

Hence, with increasing atomic number, the enhanced nuclear charge leads to contraction in the size of atoms and ions.

(iv) The atomic volumes of Europium and Ytterbium are unexceptely large. The large atomic size of Eu and Yb suggest weaker bonding in the solid elements. Both these elements have only two electrons extra than the stable configurations (half filled, f^7, and completely filled, f^{14}), hence they utilise two electrons in metallic bonding as in the case with barium.

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Effects of Lanthanide Contraction :

Close resemblance of Lanthanides : The general decrease in the sizes of the lanthanides with an increase in their nuclear charges result in a small increase in their ionisation energies. Hence their basic and ionic nature gradually decreases from La to Lu.This also explains the variations in properties such as increased tendency for hydrolysis and formation of complex salts and decreased thermal stability, solubility of their salts.

Similarity of Yttrium with lanthanides : The properties of Yttrium are so similar to the lanthanides that it is considered more a member of the lanthanide series than a congener of scandium.

Anomalous behaviour of post-lanthanides : The following anomalies may be observed in the behaviour of post-lanthanide elements.

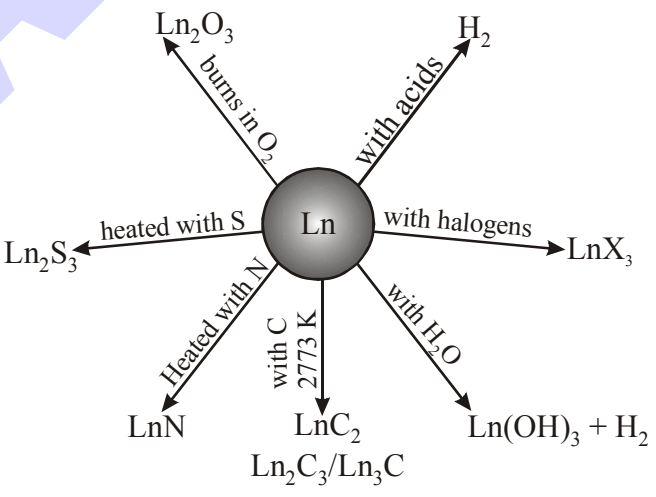
(a) **Atomic size :** The ionic radii of Zr^{+4} is about 9% more than Ti^{+4}. Similar trend is not maintained on passing from the second to third transition series. The ionic radius of Hf^{+4}, instead of increasing (because of inclusion of one more electronic shell), decreases (or is virtually equal to Zr^{+4}) as a consequence of the lanthanide contraction.This explains the close similarities between the members of the second and third transition series than between the elements of the first and second series.

(b) **Ionisation potential and electronegativity :** The effect of lanthanide contraction is also seen in the increase in the ionisation potential values and electronegativities of the elements of the third transition series, contrary to the general trend.Because of the lanthanide contraction, the post-lanthanide elements have stronger positive field and thus the electrons are held more tightly.

The greater effective nuclear charge of the former make them more electronegative than the latter.

(c) **High density :** Because of lanthanide contraction, the atomic sizes of the post lanthanide elements become very small. Consequently, the packing of atoms in their metallic crystals become so much compact that their densities are very high.The densities of the third transition series elements are almost double to those of the second series elements.

(d) **Chemical reactions of the lanthanoids.**



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Application of lanthanides :

Cerium is most useful element in the lanthanides –

- Ceramic application – CeO_2, La_2O_3, Nd_2O_3 and Pr_2O_3 are used as decolourizing agents for glasses.
- CeS (m.p. – 2000^0C) is used in the manufacture of a speciel type of crucibles and refractories.
- Lanthanide compounds like cerium molybdate, cerium tungstate are used as paints and dyes.
- In textile and leather industries (Ce salts).
- Mish metal is pyropheric and is used in cigarette & gas lighter.

ACTINIDES (5f - BLOCK ELEMENTS)

(i) The elements in which the extra electron enters 5f-orbitals of (n – 2)th main shell are known as actinides.

(ii) The man-made eleven elements Np_{93} – Lr_{103} are placed beyond uranium in the periodic table and are collectively called trans-uranium elements.

(iii) Th, Pa and U first three actinides are natural elements.

Electronic configuration :

The general configuration of actinides may be given as $5f^{1-14}\ 6d^{0-1},7s^2$.

Atomic No.	Elements	Symbol	Electronic Configuration
90	**Thorium**	**Th**	**$5f^0 6d^2 7s^2$**
91	Protactinium	Pa	$5f^2 6d^1 7s^2$
92	**Uranium**	**U**	**$5f^3 6d^1 7s^2$**
93	**Neptunium**	**Np**	**$5f^4 6d^1 7s^2$**
94	**Plutonium**	**Pu**	**$5f^6 6d^0 7s^2$**
95	Americium	Am	$5f^7 6d^0 7s^2$
96	**Curium**	**Cm**	**$5f^7 6d^1 7s^2$**
97	Berkelium	Bk	$5f^9 6d^0 7s^2$
98	Californium	Cf	$5f^{10} 6d^0 7s^2$
99	Einstenium	Es	$5f^{11} 6d^0 7s^2$
100	Fermium	Fm	$5f^{12} 6d^0 7s^2$
101	Mandelevium	Md	$5f^{13} 6d^0 7s^2$
102	**Nobelium**	**No**	**$5f^{14} 6d^0 7s^2$**
103	**Lawrencium**	**Lr**	**$5f^{14} 6d^1 7s^2$**

Oxidation states :

(i) In lanthanides and actinides +3 oxidation is the most common for both of the series of elements.

(ii) This oxidation state becomes increasingly more stable as the atomic number increases in the actinide series.

(iii) Highest oxidation states in the actinides is +7 exhibited by Np_{93} & Pu_{94}, it is unstable.

(iv) Highest stable oxidation state is +6 shown by U_{92}.

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Other Properties :

- **Physical appearance :** Acitinides are silvery white metals. They get tarnished when exposed to the attack of alkalies.
- **Density :** All the actinides except **thorium** and **americium** have high densities.
- **Colour :** Actinide ions are generally coloured. The colour of actinide ions depends upon the number of 5f-electrons. The ions containing no unpaired 5f-electrons (exactly full filled f-subshell) are colourless, as expected.
- **Ionisation energies :** Ionisation energies values of actinides are low.
- **Electropositive character :** All the known actinide metals are **highly electropositive**. They resemble lanthanide series in this respect.
- **Melting Boiling properties :** They have **high melting and boiling points.** They do not follow regular gradation of melting or boiling points with increase in atomic number.
- **Magnetic properties :** The actinide elements are paramagnetic due to the presence of unpaired electrons.
- **Radioactive nature :** All the actinides are radiaoactive in nature.
- **Actinide contraction :** The size of atom/cation decrease regularly along the actinides series. The steady decrease in ionic radii with increase in atomic number is referred to as **actinide contraction.** This is due to poor shielding of 5f-electrons.

Comparision of lanthanides and Actinides

Points of Resemblance :

(i) Both lanthanides and actinides show a dominant oxidation state of +3.

(ii) Both are electropositive and act as strong reducing agents.

(iii) Cations with unpaired electrons in both of them are paramagnetic.

(iv) Most of the cations of lanthanides and actinides are coloured.

(v) Both of them show a steady decrease in their ionic radii along the series. Thus, lanthanides show **lanthanide contraction** and actinides show **actinide contraction.**

***Difference between lanthanides & Actinides* :**

	Lathanides		Actinides
1.	Besides the most common oxidation state of +3 lanthanides show +2 and +4 oxidation states in case of certain elements.	1.	Besides the most common oxidation state of +3, actinides show +4, +5 and +6 oxidation states in case of certain elements.
2.	Lanthanides have less tendency towards complex formation.	2.	Actinides have a stronger tendency towards complex formation.
3.	Except promethium, they are non radioactive.	3.	All the actinides are radioactive.
4.	Oxides and hydroxide of lanthanides are less basic.	4.	Oxides and hydroxides of actinides are more basic

Some important uses of actinides are as follows –

Thorium : Thorium is used in atomic reactors as fuel rods and in the treatment of cancer.

Uranium : Uranium is used as nuclear fuel. Its salts are used in glass industry (for imparting green colour). textile industry and also in medicines.

Plutonium : Plutonium is used as fuel for atomic reactors as well as in atomic bombs.

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